

Module 7: Cost, Revenue, and Profit — Answer Key

Prequiz

1. $F = 180$, $v = 6$, $p = 18$. Thus

$$C(q) = \boxed{180 + 6q}, \quad R(q) = \boxed{18q}, \quad \pi(q) = \boxed{12q - 180}.$$

- 2.

$$C(25) = 180 + 6(25) = \boxed{330},$$

$$R(25) = 18(25) = \boxed{450},$$

$$\pi(25) = 450 - 330 = \boxed{120}.$$

The business has a profit.

- 3.

$$q^* = \frac{180}{18 - 6} = \boxed{15}.$$

At $q = 14$, profit is -12 ; at $q = 16$, profit is $+12$.

4. Marginal cost = $\boxed{4}$, marginal revenue = $\boxed{9}$, marginal profit = $\boxed{5}$. Break-even:

$$q^* = \frac{250}{9 - 4} = \boxed{50}.$$

Ten additional meals after break-even add $10(5) = \boxed{\$50}$ profit.

5. To the right of break-even, **revenue is larger than cost**, so profit is positive.

Matching quiz

- 1.

$$C(q) = \boxed{240 + 3q}, \quad R(q) = \boxed{9q}, \quad \pi(q) = \boxed{6q - 240}.$$

2. At $q = 50$:

$$C = \boxed{390}, \quad R = \boxed{450}, \quad \pi = \boxed{60}.$$

- 3.

$$q^* = \frac{240}{9 - 3} = \boxed{40}.$$

The minimum whole number for nonnegative profit is 40.

4. Marginal profit = $15 - 7 = \boxed{8}$. Twelve additional units add $12(8) = \boxed{\$96}$ profit.

5. Revenue is higher than cost, so profit is positive.